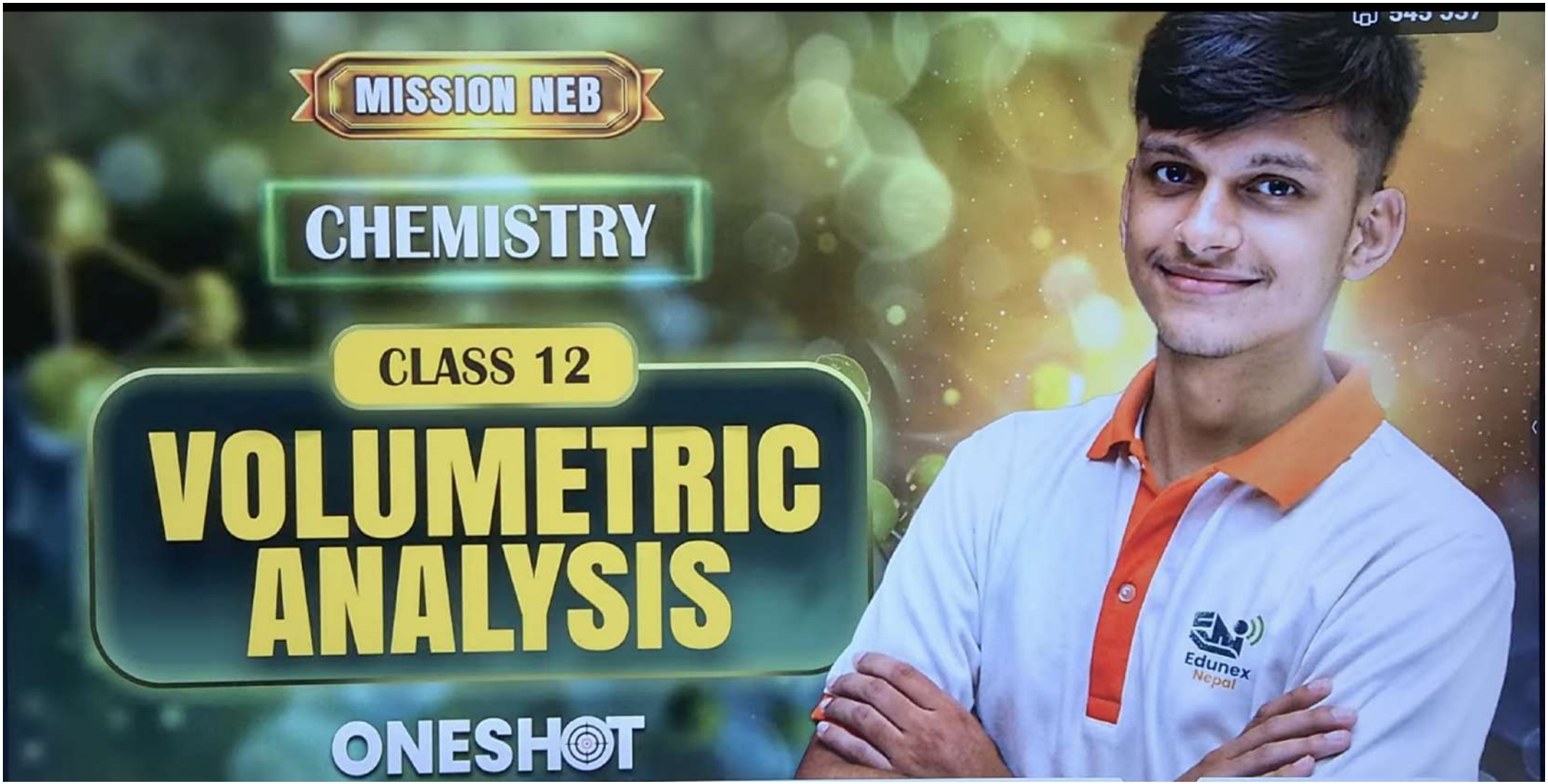
MISSION NEB
CHEMISTRY
CLASS 12
VOLUMETRIC ANALYSIS
ONESHOT
Edunex
Nepal

1. Equivalent weight.

$$\text{Eq wt} = \frac{\text{Atomic wt / Molecular wt}}{\text{n-factor}}$$

a) Acid → Basicity

b) Base → acidity

c) Salt → Total no of -ve charge / Total no of -ve charge.

$$E_{HCl} = \frac{\text{Mol. wt}}{\text{n factor}}$$

$1 + 35.5 = 36.5$

$$= \frac{\text{Mol. wt}}{\text{Basicity}}$$

Basicity → no of replacable H^+ ions.

$$= \frac{\text{Mol. wt}}{1}$$

$$= \frac{36.5}{1}$$

$$= 36.5$$

$$E_{H_2SO_4} = \frac{98}{2}$$

$1 \times 2 + 32 + 16 \times 4 = 98$

$= 49$

a) $E_{H_3PO_4} = \frac{\quad}{3}$ phosphoric acid

b) $E_{H_3PO_3} = \frac{\quad}{2}$ phosphorus acid.

c) $E_{H_3PO_2} = \frac{\quad}{1}$ Hypophosphorus acid

@ no of oxygen atom − 1

$= 4 - 1 = 3$

$E_{NaOH} = \frac{\text{Mol. wt}}{\text{n factor}}$

$23+17 = 40$

$= \frac{40}{\text{acidity}}$ → no of replacable OH^- ion

$= \frac{40}{1} = 40$

$E_{Mg(OH)_2} = \frac{58}{2} =$

$24+(16+1)\times 2$

$24+17\times 2$

$24+34$

$E_{Al_2(SO_4)_3} = \frac{\quad}{6}$

Salt $= \frac{\quad}{\quad}$

$Al(OH)_3 + H_2SO_4$

* Salt:

$E_{Na_2CO_3} = \frac{\text{Mol. wt}}{\text{n factor}} = \frac{\text{Mol wt}}{2} = 53$

↳ Total +ve charge

$Na_2CO_3 \longrightarrow 2Na^+ + CO_3^{--}$

$A_xB_y \rightleftharpoons xA^{y+} + yB^{x-}$

$Na_2(CO_3)_1 \rightleftharpoons 2Na^{1+} + CO_3^{2-}$

$Al_2(SO_4)_3 \rightleftharpoons 2Al^{3+} + 3SO_4^{2-}$

n factor = 6

2. Equivalent weight of $KMnO_4$ (oxidizing agent).

↳ change in O.NO

1) Acidic Medium :

$$K\overset{+7}{\underline{Mn}}O_4 + H_2SO_4 \longrightarrow K_2SO_4 + \overset{+2}{\underline{Mn}}SO_4 + H_2O + [O]$$

$x - 2 = 0$

$\therefore x = +2$

or, $+1 + x + (-2) \times 4 = 0$

or, $+1 + x - 8 = 0$

$\therefore x - 7 = 0 = +7$

change in ONO = 5

n factor = 5

$$\text{Eq wt} = \frac{158}{5} = _$$

* Basic Medium:

$$K\overset{+7}{Mn}O_4 + KOH \longrightarrow K_2\overset{+6}{Mn}O_4 + H_2O + [O]$$

$+2 + x - 8 = 0$

$\therefore x = +6$

@ B A N

change in = 1 5 3

change in ONO = 1

$$Eq\ wt = \frac{158}{1} = 158$$

* Neutral Medium:

$$K\overset{+7}{Mn}O_4 + H_2O \longrightarrow KOH + \overset{+4}{Mn}O_2 + [O]$$

$x + (-2) \times 2 = 0$

$\therefore x = +4$

change in ONO = 3

$$\therefore Eq\ wt = \frac{158}{3} = -$$

3. What is the equivalent weight of H_3PO_4 in the following reactions?

a) $2\ NaOH + H_3PO_4 \rightarrow Na_2HPO_4 + 2H_2O$

b) $Ca(OH)_2 + H_3PO_4 \rightarrow CaHPO_4 + 2H_2O$

n factor = ?

① H

② → Replacable H^+ ion = n factor.

a) $2\ NaOH + H_3PO_4 \longrightarrow Na_2HPO_4 + 2H_2O$

H_3PO_4 → Acid → replacable ③

Na_2HPO_4 → ① H → Remaining

② H → Replacable

$$\text{Eq wt} = \frac{H_3PO_4}{2}$$

$$= \frac{1\times3+31+16\times4}{2}$$

$$= \frac{98}{2} = 49$$

4. Concentration terms:

a) Percentage solution

$$\%\,\frac{w}{v} = \frac{\text{weight of solute (g)}}{\text{Vol of sol}^n\text{ (ml)}} \times 100\%$$

Eg: 20% $\frac{w}{v}$ of NaOH:

wt of solute = 20g

$$\frac{\text{Solute}}{\text{Solution}}$$

$$\%\,\frac{w}{v} = 20\% = \frac{20}{100} \times 100\%$$

Vol of sol^n = 100 ml

$$= 20\%$$

$$* \ \% \frac{w}{w} = \frac{\text{weight of Solute (g)}}{\text{weight of Solution (g)}} \times 100\%$$

Eg: 20% $\frac{w}{w}$ Urea

$$20\% = \frac{20}{100} \times 100$$

$= 20\%$

wt of solute = 20g

wt of solution = 100g

5. Gram per litre (g/L).

$$g/L = \frac{\text{wt of solute (g)}}{\text{vol of sol}^n\ (L)}$$

6. Normality.

- Normality = $\frac{\text{no of g eq of solute}}{\text{Vol of sol}^n \text{ (ltr)}}$

- No. of g eq of solute = $\frac{\text{weight of solute}}{\text{Equivalent wt}}$

- Equivalent wt = $\frac{\text{Mol wt}}{\text{n factor}}$

N = Normal solution

$\frac{N}{2}$ = Seminormal "

$\frac{N}{5}$ = pentinormal "

$\frac{N}{10}$ = Decinormal "

$\frac{N}{100}$ = centinormal "

7. Molarity.

$$\text{Molarity} = \frac{\text{no of moles of solute}}{\text{vol of sol}^n\ (l)} \quad (M)$$

$$\text{no of moles} = \frac{\text{Weight}}{\text{Molecular wt}}$$

$$N = \frac{\text{No of g eq}}{\text{V of sol}^n\ (l)}$$

$$\text{or, } N = \frac{\frac{w}{E}}{V\ (l)}$$

$$\boxed{\therefore w = NEV_{ltr}}$$

$$\therefore w = \frac{NEV_{ml}}{1000}$$

w = Given wt

N = Normality

E = Eq wt

V = Vol of sol^n

$$g/L = \frac{\text{wt of solute}}{\text{vol of sol}^n\ (l)}$$

g/L = Normality × Eq wt — (I)

g/L = Molarity × Mol wt — (II)

$N \times$ Eq wt $= M \times$ Mol·wt

$$\text{or } N \times \frac{\cancel{\text{Mol wt}}}{n \text{ factor}} = M \times \cancel{\text{mol wt}}$$

$\therefore$ Normality = Molarity × n factor

8. Molality.

$$\text{Molality} = \frac{\text{no of moles of solute}}{\text{wt of solvent (kg)}}$$

(m)

m

$\frac{m}{2}$

$\frac{m}{5}$

$\frac{m}{10}$

$\frac{m}{100}$

9. Mole fraction.

Solute

Solvent

Solution = solute + solvent

$$\chi_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

$$\chi_{solvent} = \frac{n_{solvent}}{n_{solute} + n_{solvent}}$$

$$\therefore \chi_{solute} + \chi_{solvent} = \frac{n_{solute}}{n_{solute} + n_{solvent}} + \frac{n_{solvent}}{n_{solute} + n_{solvent}} = 1$$

a) <1

b) >1

c) 0

d) 1

10. PPM (parts per million).

$$10^6$$

$$\frac{w}{v} = \frac{\text{weight of solute}}{\text{vol of sol}^n\ (ml)} \times 10^6$$

$$\frac{w}{w} = \frac{\text{weight of solute}}{\text{weight of sol}^n} \times 10^6$$

11. PPB (parts per billion).

$$\frac{w}{v} = \frac{\text{weight of solute}}{\text{vol of soln (ml)}} \times 10^9$$

$$\frac{w}{w} = \frac{\text{weight of solute (g)}}{\text{weight of soln (g)}} \times 10^9$$

12. Convert the following:

a) 2.5 M H_3PO_4 into normality

b) 4.9 M H_2SO_4 into gram per litre

a) 2.5M → N = ?

Normality = Molarity × n factor

= 2.5 × 3

= 7.5N

H_3PO_4 = 3

b) Soln

M = 4.9

g/L = ?

g/L = M × Molwt

= 4.9 × 98

= 480.2 g/L

13. What is the normality of 20 cc of 2 M phosphoric acid?

Soln

phosphoric acid = H_3PO_4

Molarity = 2

Normality = Molarity x n factor

= 2 x 3

= 6N

14. What mass of Na_2CO_3 is required to prepare 50 cc of its seminormal solution?

Sol^n

$$w = \frac{N E V_{ml}}{1000}$$

$$\text{or, } w = \frac{0.5 \times 53 \times 50}{1000}$$

$$\therefore w = 1.325g$$

Seminormal $= \frac{1}{2} N$

$= 0.5N$

Eq wt of $Na_2CO_3 = \frac{\text{Mol wt}}{\text{n factor}}$

$= \frac{106}{2}$

$= 53$

15. If a student prepares a solution by dissolving 5.3 g of Na_2CO_3 in 500 ml of solution, what will be the molar concentration of the solution?

Molarity = ?

wt of Na_2CO_3 = 5.3g

vol of sol^n = 500 ml

$N = M \times n\text{factor}$

$= 0.1 \times 2$

$= 0.2N$

$$\text{Molarity} = \frac{\text{no of moles of solute}}{\text{Vol of sol}^n\ (\text{ltr})}$$

$$w = \frac{NEV}{1000}$$

N → M

$$= \frac{\frac{w}{\text{Mol wt}}}{\frac{500}{1000}} = \frac{\frac{5.3}{106}}{\frac{1}{2}} = 0.1M$$

16. Which one has higher concentration?

a) 80 g/L NaOH solution and 3 M NaOH

b) 5.3 g/L Na_2CO_3 and N/10 Na_2CO_3 solution

M → 0.1N

Sol^n

(a) 80 g/L NaOH = 2M NaOH

g/L = Molarity × Mol·wt

or, $M = \frac{g/L}{Mol\ wt}$

or, $M = \frac{80}{40}$

∴ M = 2M

NaOH

= 23 + 16 + 1

= 40

∴ 3M NaOH has higher Conc^n

(b) g/L = N × Eq wt

or, 5.3 = N × 53

∴ N = 0.1N Na_2CO_3

$\frac{N}{10}$ Na_2CO_3

= 0.1N Na_2CO_3

∴ Both has same concentration.

17. Calculate molality of one litre of 93% H_2SO_4 solution (wt by volume) if the density of the solution is 1.84 g/ml.

wt of solute + wt of solvent = wt of sol^n

Sol^n

$$\text{Molality} = \frac{\text{no of moles of solute}}{\text{wt of solvent (kg)}}$$

Given,

93·1·H_2SO_4 (w/v) = 93 g of H_2SO_4 is present in 100 ml of sol^n.

wt of solute = 93 g
vol of sol^n = 100 ml

100 ml of sol^n → 93 g of H_2SO_4

1 L = 1000 ml of sol^n → $\frac{93}{100} \times 1000$ g of H_2SO_4 = 930 g

18. A solution of HCl contains 38% HCl by mass. Find the molarity of this solution if the density of the solution is 1.19 g/cc.

Soln

$\% \frac{w}{w}$ = percentage by mass = 38%

M = ?

$\rho = 1.19$ g/cc

⇓

Relative density

= Specific gravity

$$M = \frac{\% \frac{w}{w} \times \text{Sp. gravity} \times 10}{\text{Molecular wt}}$$

$$= \frac{38 \times 1.19 \times 10}{36.5} = 12.38\ M$$

$$N = \frac{\% \frac{w}{w} \times \text{sp gravity} \times 10}{\text{Eq wt}}$$

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19. Calculate the molarity of 5% H_2SO_4 solution.

Soln

5% H_2SO_4 (w/v)

= 5 g of H_2SO_4 is present in 100 ml of soln

wt of solute = 5g

Vol of soln = 100ml

$$M = \frac{\text{no of moles}}{\text{Vol of soln (ltr)}}$$

$$= \frac{\frac{w}{M}}{\frac{100}{1000}}$$

$$= \frac{5}{98} \times \frac{1000}{100}$$

$$= 0.51M$$

20. A solution contains 1 mole of alcohol and 4 moles of water. What will be the mole fraction of water and alcohol?

Soln

$n_{alcohol} = 1$ mole

$n_{water} = 4$ moles

$$\chi_{water} = \frac{n_{water}}{n_{water} + n_{alcohol}} = \frac{4}{4+1} = \frac{4}{5}$$

$$\chi_{alcohol} = \frac{n_{alcohol}}{n_{alcohol} + n_{water}} = \frac{1}{1+4} = \frac{1}{5}$$

Q21. Principle of volumetric analysis.

$A + B \longrightarrow$ product

↳ complete Rxn

∴ No. of g eq of A = No. of g eq of B

$N = \frac{\text{No. of g eq of solute}}{\text{Vol of soln (l)}}$

∴ $N \times V(l) =$ No. of g eq

Acid + Base $\longrightarrow$ Salt + H_2O

↓

complete Rxn

⇓

Neutralization.

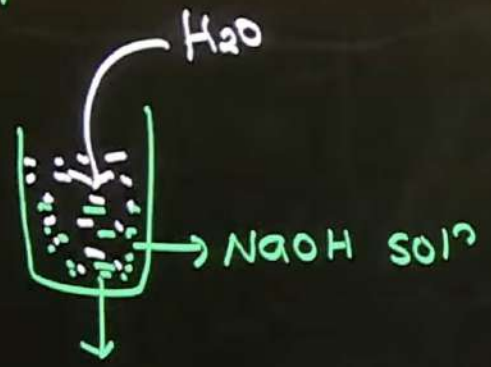
No. of g eq of Acid = No. of g eq of Base

or, $(\text{Normality})_{Acid} \times (\text{Volume})_{Acid} = (\text{Normality})_{Base} \times (\text{Volume})_{Base}$

or, $N_A \times V_A = N_B V_B$ ∴ $N_1V_1 = N_2V_2$

Q22. What volume of water should be added to 50 ml of seminormal NaOH solution to make it exactly decinormal?

Solⁿ



$V_1 = 50ml$

$N_1 = \frac{1}{2}N$

$V_2 = ?$

$N_2 = \frac{1}{10}N$

Now,

$N_1V_1 = N_2V_2$

or, $0.5 \times 50 = 0.1 \times V_2$

$\therefore V_2 = 250ml$

$\therefore$ vol of water added $= (250 - 50)ml$

$= 200ml$

Q23. How many ml of concentrated HNO3 of specific gravity 1.41 containing 69% by mass are required to prepare 500 ml of 0.5 N HNO3?

Sol^n

vol of conc HNO_3 = ?

Given,

Sp. gravity = 1.41

$\% \frac{w}{w} = 69$

$N_2 = 0.5N$

$V_2 = 500ml$

Now,

$$\text{Normality } (N_1) = \frac{\% \frac{w}{w} \times \text{sp gravity} \times 10}{\text{Eq wt}}$$

$$= \frac{69 \times 1.41 \times 10}{63/1}$$

$$= 15.33 N$$

Now,

$N_1V_1 = N_2V_2$

or, $15.33 \times V_1 = 0.5 \times 500$

$\therefore V_1 = 16.188 ml$

Q24. 100 cc of N/2 HCl, 30 cc of N/10 HNO3 and 60 cc of N/5 H2SO4 are mixed. What is the normality of the mixture?

Conditions:

1) If the mixture of acids are given,

$$N_R = \frac{N_1V_1 + N_2V_2 + \cdots\cdots}{V_1 + V_2 + \cdots\cdots}$$

2) If the mixture of bases are given,

$$N_R = \frac{N_1V_1 + N_2V_2 + \cdots\cdots}{V_1 + V_2 + \cdots\cdots}$$

3) For the mixture of acids and bases,

$$N_R = \frac{\overset{A}{N_1V_1} - \overset{B}{N_2V_2}}{V_1 + V_2}$$

If N_1V_1 = (Acid)
N_2V_2 = (Base)

N_R = +ve (Resulting sol^n = acidic)
N_R = −ve (Resulting sol^n = Basic)

HCl

$V_1 = 100\,cc$

$N_1 = \frac{N}{2}$

HNO_3

$V_2 = 30\,cc$

$N_2 = \frac{N}{10}$

H_2SO_4

$V_3 = 60\,cc$

$N_3 = \frac{N}{5}$

$$N_R = \frac{N_1V_1 + N_2V_2 + N_3V_3}{V_1 + V_2 + V_3}$$

$$= \frac{0.5 \times 100 + 0.1 \times 30 + 0.2 \times 60}{100 + 30 + 60}$$

$$= 0.34N$$

Q25. What volume of $\frac{N}{2}$ and $\frac{N}{10}$ HCl must be mixed to give 2 litres of $\frac{N}{5}$ HCl?

Soln

HCl

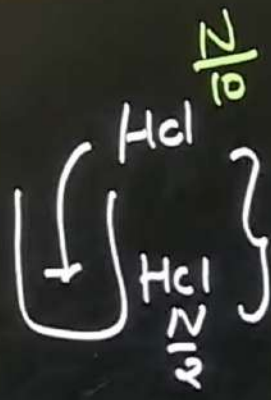
$V_1 = x$

$N_1 = \frac{N}{2}$

HCl

$V_2 = 2 - x$

$N_2 = \frac{N}{10}$



$\rightarrow N_R = \frac{N}{5}$

$V_R = 2$ litres

$V_1 + V_2 = V_R = 2$ litres

Now,

$$N_R = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

$$\text{or, } \frac{1}{5} = \frac{\frac{1}{2} \times x + \frac{1}{10} \times (2-x)}{2}$$

$$\text{or, } 0.2 = \frac{0.5x + 0.1(2-x)}{2}$$

$\therefore x = 0.5$ litres.

$\therefore V_2 = 2 - x$

$= 2 - 0.5$

$= 1.5$ litres

Q26. x g of Na2CO3 reacts completely with 50 ml of 2 N HCl. Find the value of x.

Sol^n

$$Na_2CO_3 + HCl$$

complete Rxn:

No. of g eq of Na_2CO_3 = No. of g of HCl

$$\text{or, } \left(\frac{w}{Eq\,wt}\right)_{Na_2CO_3} = \left(\frac{N \times V_{ml}}{1000}\right)_{HCl}$$

$$\text{or, } \frac{x}{53} = \frac{2 \times 50}{1000}$$

$$\therefore x = 5.3\,gm$$

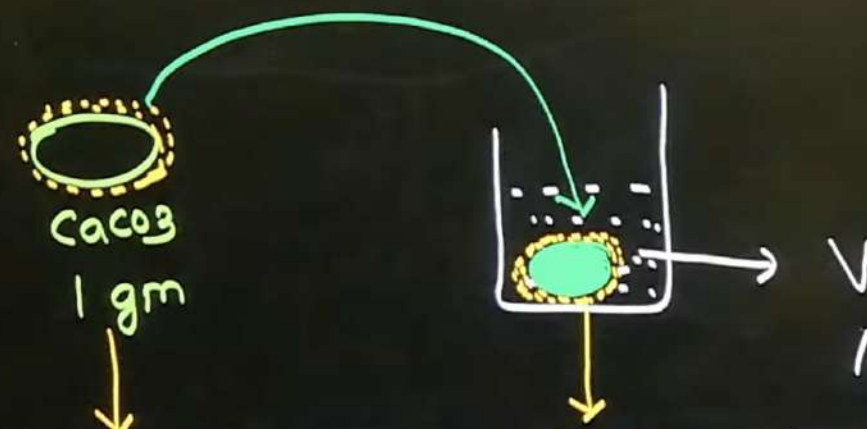
$$N = \frac{\text{No. of g eq}}{\text{vol of sol}^n(l)}$$

① ∴ No. of g eq = $N \times V(l)$

② No. of g eq = $\frac{w}{Eq\,wt}$

. 1 g of ordinary sample of limestone is dissolved in 16.6 cc of 0.92 N HCl solution leaving some sandy residue. Find the percentage purity of the sample.

Sol



$V_{HCl} = 16.6\ cc$

$N_{HCl} = 0.92N$

wt of impure CaCO3 = 1 gm

pure part ही → React

% purity = $\frac{\text{wt of pure sub}}{\text{wt of impure sub}} \times 100\%$

$CaCO_3 + HCl$

No of g eq of pure $CaCO_3$ = No. of g eq of HCl

or, $\left(\frac{w}{E}\right)_{pure\ CaCO_3} = \left(\frac{N \times V_{ml}}{1000}\right)_{HCl}$

$Eq = \frac{100}{2} = 50$

or, $\frac{w_{pure\ CaCO_3}}{50} = \frac{0.92 \times 16.6}{1000}$

$\therefore w_{pure\ CaCO_3} = 0.76g$

Now,

$\% \text{ purity} = \frac{\text{wt of pure sub}}{\text{wt of impure sub}} \times 100\%$

$= \frac{0.76}{1} \times 100\%$

$= 76\%$

- $A + B \longrightarrow P$

Complete Rxn:

No. of g eq of A = No. of g eq of B

- $A + X$ (B)

A → complete

B → excess

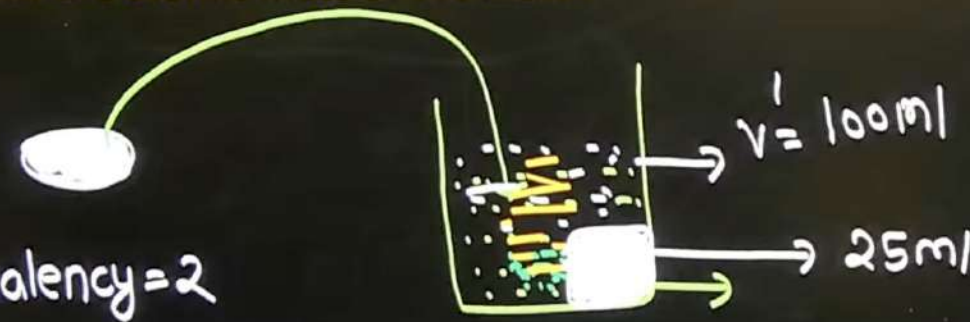
$\cancel{A} + B\uparrow$

$B \longrightarrow A + X$

No. of g eq of B = No. of g eq of A + No. of g eq of X

Q28. 0.4 g of a divalent metal was dissolved in 50 cc of 0.64 N HCl and the solution was diluted to 100 cc. 25 cc of this solution required 27.3 cc of 0.11 N NaOH for complete neutralization. Find the atomic mass of the metal.

Soln



$V' = 100ml$

25ml of the solution is neutralized by 27.3 cc of 0.11N NaOH

valency = 2

wt of metal = 0.4g

HCl, V = 50 ml

N = 0.64N

Now,

No. of g eq of Acid = No. of g eq of metal + No. of g eq of base

or, $\left(\frac{N \times V_{ml}}{1000}\right)_{Acid} = \left(\frac{w}{E}\right)_{metal} + \left(\frac{N \times V_{ml}}{1000}\right)_{Base}$

or, $\frac{0.64 \times 50}{1000} = \frac{0.4}{E} + \frac{0.11 \times 109.2}{1000}$

· 100 ml of the soln is neutralized by ·

$\frac{27.3}{25} \times 100ml \Rightarrow 109.2ml$ 0.11N NaOH

$\therefore E = 20g$

Now,

$$\text{Equivalent wt} = \frac{\text{Mol wt}}{\text{Valency}}$$

$$\text{or, } 20 = \frac{\text{Mol wt}}{2}$$

$\therefore$ Mol wt = 40g //

Q29. 4 g of a divalent metal was dissolved in 100 cc of 2 M H_2SO_4. The excess acid required 30 cc of 1 N NaOH for complete neutralization. Find the atomic mass of the metal.

$N = M \times n = 2 \times 2 = 4$

Sol^n

O

wt = 4g

Valency = 2

NaOH → $V_b = 30cc$

$N_b = 1N$

H_2SO_4

M = 2M

V = 100ml

No. of g eq of H_2SO_4 = No. of g eq of Metal + No. of g eq of NaOH

Or, $\left(\frac{N \times V_{ml}}{1000}\right)_{H_2SO_4} = \left(\frac{w}{E}\right)_{Metal} + \left(\frac{N \times V_{ml}}{1000}\right)_{NaOH}$

or $\frac{4 \times 100}{1000} = \frac{4}{E} + \frac{1 \times 30}{1000}$

$\therefore E = 10.81 g$

$\therefore$ At wt = Eq wt × n factor

$= 10.81 \times 2$

$= 21.62 g$ //

Homework:

9) x g of a metal (Eq wt = 12) was completely dissolved in 100 cc of $\frac{N}{2}$ HCl. The volume was then made upto 500 cc. 25 cc of this diluted acid required 17.5 cc of $\frac{N}{10}$ NaOH for complete neutralization. Find the value of x.

Q30. 0.715 g of $Na_2CO_3 \cdot xH_2O$ required 20 ml of seminormal HCl solution for complete neutralization. Find the value of x.

Sol^n

$Na_2CO_3 \cdot xH_2O$ vs HCl $\quad N = \frac{1}{2} = 0.5N$

No. of g eq of $Na_2CO_3 \cdot xH_2O$ = No. of g eq of HCl

or, $\left(\frac{w}{E}\right)_{Na_2CO_3 \cdot xH_2O} = \frac{0.5 \times 20}{1000}$

or, $\frac{0.715}{53+9x} = \frac{0.5 \times 20}{1000}$

$\therefore x \approx 2$ //

$Na_2CO_3 \cdot 2H_2O$

$E_{Na_2CO_3 \cdot xH_2O}$

$= \frac{\text{Mol wt}}{\text{n factor}}$

$= \frac{106 + 18x}{2}$

$= \frac{106}{2} + \frac{18x}{2}$

$= 53 + 9x$

Q31. 12 g of commercial zinc reacts with excess dilute H_2SO_4. The total volume of H_2 gas produced is 4.2 litres at 570 mm Hg pressure and 279 K temperature. Find the percentage purity of zinc.

Sol^n

$$Zn + \text{dil } H_2SO_4 \longrightarrow ZnSO_4 + H_2$$

	(limiting)	(excess)		
	1 mole	1 mole	1 mole	1 mole
	→ 65g (pure wt)			?

• 1 mole H_2 is produced from 65g of Zn.

• 760 mm of Hg = 1 atm ∴ 570 mm Hg ⇒ 0.75 atm

$R = 0.0821$ litre atm mole^{-1} kelvin^{-1}

(U.G.C)

$$\% \text{ purity} = \frac{\text{wt of pure...}}{\text{wt of impure...}} \times 100\%$$

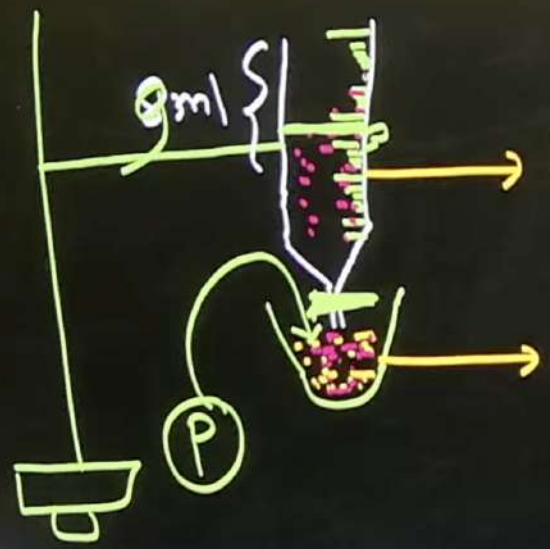
$$PV = nRT$$

$$\text{or, } n = \frac{PV}{RT}$$

$$\text{or, } n = \frac{0.75 \times 4.2}{0.0821 \times 279}$$

$$=$$

❖ Titration



HCl vs NaOH

Burette → Top → Titrant

conical flask (NaOH) (Analyte) → Titrand @down

$V_{NaOH} = 10ml$

$N_{NaOH} = ?$

$V_{HCl} = ?$ 8ml

$N_{HCl} = \frac{N}{10}$

$10 \times x = 8 \times 0.1$

$\therefore x =$

⇒ Indicator
- Methyl orange
- phenolpthalein

- 1 drop → pink → colourless
- Eq point → End point

- Titrant → Thaha भएको
- Titrand → determine गर्नु पर्ने

Titration error :

* Some terminologies:

1) Titrant: The solution whose concentration is known.

2) Titrand: The solution whose concentration has to be determined.

3) Indicator: The substance used in titration which inspect the completion of reaction by sharp change in colour of solution.

4) Equivalence point: The point during the titration process where no of g eq of unknown solution is equal to the no of g eq of known solution.

5) End point: The point at which the reaction is completed during the titration that is indicated by change in colour.

6) Titration error: The difference between end point and equivalence point is Titration error. (volume):

* Acidimetry:

Acid की measurement → unknown ⇒ Acid
→ known → Alkali

Trigonometry

* Alkalimetry:

Base की measurement → Unknown ⇒ Alkali
→ known ∴ Acid

* Redox Titration:

$KMnO_4 + H_2SO_4 + H_2C_2O_4 \longrightarrow K_2SO_4 + MnSO_4 + CO_2 + H_2O$

$KMnO_4$ → Oxidising agent

$H_2C_2O_4$ → Reducing agent.

O·NO

$(COOH)_2$

Redox Reaction

Reduction oxidation → oxidizing agent / → Reducing agent

Redox

· $KMnO_4$ → self indicator.

COOH
|
COOH

Oxalic acid.

weak acid.

↓

H^+ X

$H_2SO_4 \rightarrow H^+$

* Normality factor: (f)

$$f = \frac{\text{wt taken of solute}}{\text{wt to be taken}}$$

$$= \frac{\text{Actual weight}}{\text{Theoretical wt}}$$

Exact Normality = proposed normality × f

20ml of 0.1N HCl (f = 0.6)

$E \cdot N = 0.1 \times 0.6$
$= 0.6$

$$w = \frac{N E V_{ml}}{1000}$$
$= ___$

19.8ml

V = 20ml

* primary standard solution:

• Secondary standard solution

liquid

NaOH, HCl, H_2SO_4 etc.

↳ primary standard substance. Imp

→ pure and dry

→ High mol wt and Eq wt high (minimize error)

→ non-hygroscopic

→ non-deliquescent

Eg: Na_2CO_3, $CuSO_4 \cdot 5H_2O$ etc

(खौज)

wt = less

1g नुन

1.85 0.6

2. 1kg

Q. Deduce the normality equation M1V1 = M2V2. Experimental data obtained by titrating decinormal solution of oxalic acid with potassium permanganate solution are given below.

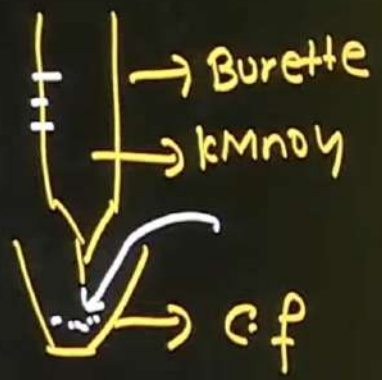
N N

Expt. No	Volume of Oxalic acid	Volume of $KMnO_4$ burrete reading (ml)	
		iintial	finalal
1	10 ml	00.0	11.5
2	10 ml	00.0	11.0
3	10 ml	00.0	11.0

→ 11.0

(a) Name the above titration. → Redox titration
(b) Calculate the equivalent weight of KMnO4.
(c) Calculate the normality of KMnO4 from the above data.
(d) Why is dilute H2SO4 added to the conical flask containing standard oxalic acid before titrating with KMnO4 solution?
(e) Identify the titrant and titrand in this titration.

Solⁿ
KMnO4 vs Oxalic Acid



= Oxalic acid.

$N_{acid} = 0.1N$

$KMnO_4$, Eq wt = $\frac{\text{Mol wt}}{\text{n factor}}$ = —